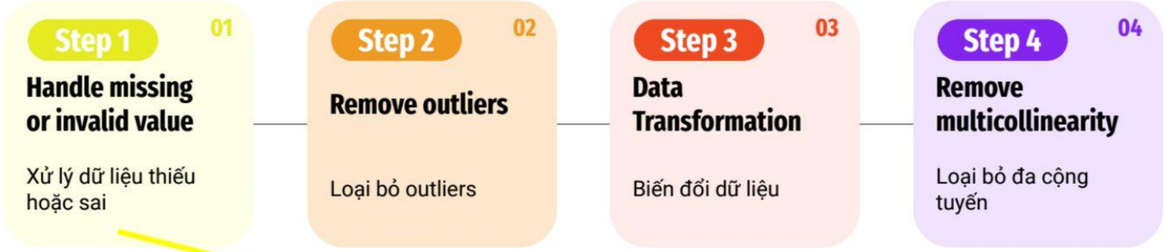
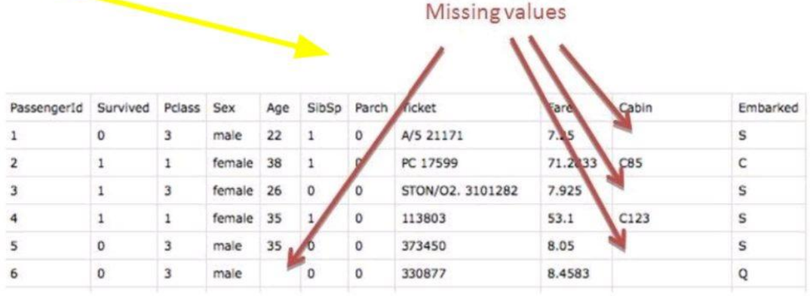
Step 3.1: Handle missing or invalid values





PassengerId	Survived	Pclass	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	male	22	1	0	A/5 21171	7.25		S
2	1	1	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	female	26	0	0	STON/O2. 3101282	7.925		S
4	1	1	female	35	1	0	113803	53.1	C123	S
5	0	3	male	35	0	0	373450	8.05		S
6	0	3	male		0	0	330877	8.4583		Q

Step 3.2: Remove outliers

Step 1 01

Handle missing or invalid value

Xử lý dữ liệu thiếu hoặc sai

Step 2 02

Remove outliers

Loại bỏ outliers

Step 3 03

Data Transformation

Biến đổi dữ liệu

Step 4 04

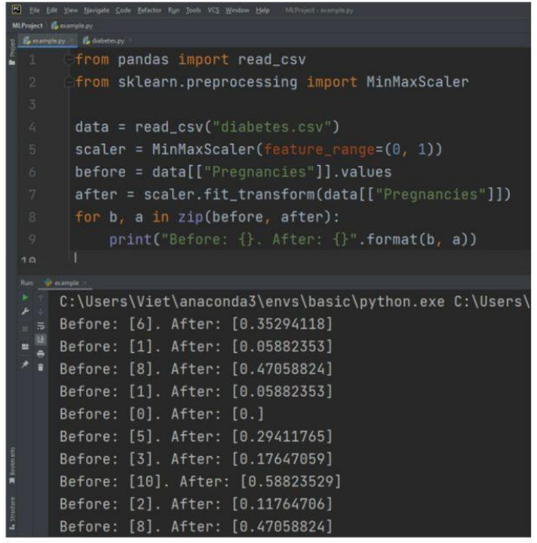
Remove multicollinearity

Loại bỏ đa cộng tuyến



Step 3.3: Tiền xử lý numerical features

MinMaxScaler

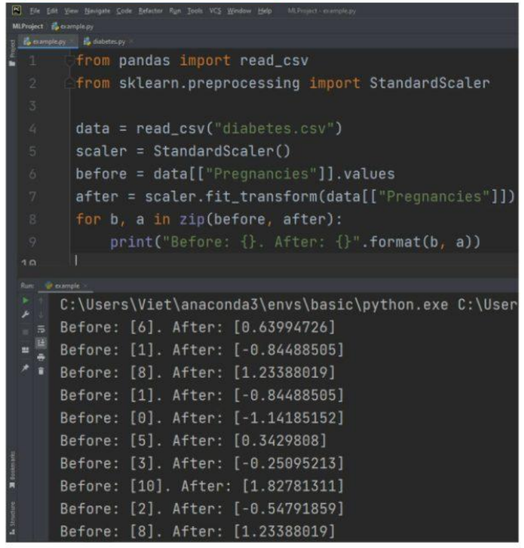


```
from pandas import read_csv
from sklearn.preprocessing import MinMaxScaler

data = read_csv("diabetes.csv")
scaler = MinMaxScaler(feature_range=(0, 1))
before = data[["Pregnancies"]].values
after = scaler.fit_transform(data[["Pregnancies"]])
for b, a in zip(before, after):
    print("Before: {}. After: {}".format(b, a))
```

```
C:\Users\Viet\anaconda3\envs\basic\python.exe C:\Users\
Before: [6]. After: [0.35294118]
Before: [1]. After: [0.05882353]
Before: [8]. After: [0.47058824]
Before: [1]. After: [0.05882353]
Before: [0]. After: [0.]
Before: [5]. After: [0.29411765]
Before: [3]. After: [0.17647059]
Before: [10]. After: [0.58823529]
Before: [2]. After: [0.11764706]
Before: [8]. After: [0.47058824]
```

StandardScaler



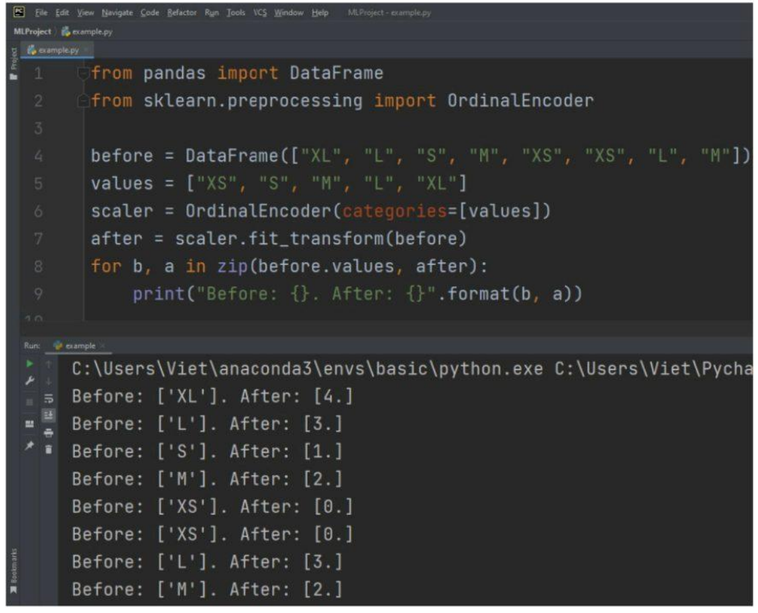
```
from pandas import read_csv
from sklearn.preprocessing import StandardScaler

data = read_csv("diabetes.csv")
scaler = StandardScaler()
before = data[["Pregnancies"]].values
after = scaler.fit_transform(data[["Pregnancies"]])
for b, a in zip(before, after):
    print("Before: {}. After: {}".format(b, a))
```

```
C:\Users\Viet\anaconda3\envs\basic\python.exe C:\User
Before: [6]. After: [0.63994726]
Before: [1]. After: [-0.84488505]
Before: [8]. After: [1.23388019]
Before: [1]. After: [-0.84488505]
Before: [0]. After: [-1.14185152]
Before: [5]. After: [0.3429808]
Before: [3]. After: [-0.25095213]
Before: [10]. After: [1.82781311]
Before: [2]. After: [-0.54791859]
Before: [8]. After: [1.23388019]
```

Step 3.3: Tiền xử lý ordinal features

OrdinalEncoder



Step 3.3: Các cách mã hóa nominal features

One-hot encoding

Color

- Red -> **1**, 0, 0, 0
- Green - > 0, **1**, 0, 0
- Blue -> 0, 0, **1**, 0
- Black -> 0, 0, 0, **1**

Advantage

- Simple

Disadvantage

- Sparse data

Hash encoding

Country

- USA -> **1, 0, 0**
- UK -> 0, 0, 1
- Korea -> 0, 1, 0
- Russia -> **1, 0, 0**

Advantage

- Simple
- Less sparse

Disadvantage

- Collision
- No inverse-mapping

Word Embedding

Animal

- Dog -> 0.27,-0.31, -0.53
- Lion -> **-0.7, 0.61, 0.42**
- Tiger -> **-0.71, 0.6, 0.38**
- Mouse -> 0.31, -0.34, 0.76

Advantage

- Memory efficient
- Relationship learnt

Disadvantage

- Word2vec model needed
- Maybe many dimensions needed

Step 3.4: Mối quan hệ giữa các variables (features, target)

Correlation

- **Mức độ 2 hay nhiều variables quan hệ tuyến tính với nhau**
- **Hệ số tương quan: [-1, 1]**
- **Correlation does not imply causation**

$$r_{xy} = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\Sigma(x_i - \bar{x})^2 \Sigma(y_i - \bar{y})^2}}$$

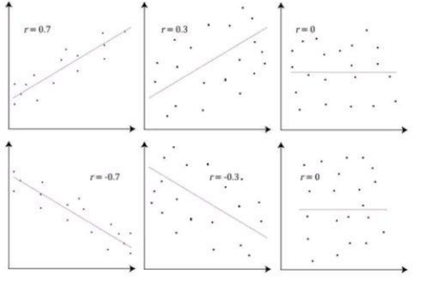
r_{xy} = correlation coefficient between x and y

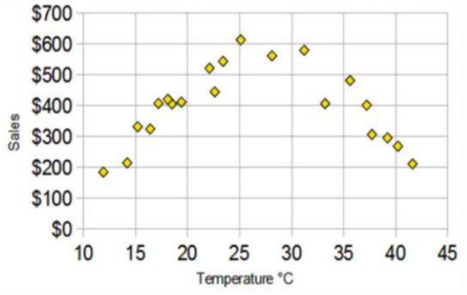
x_i = the values of x within a sample

y_i = the values of y within a sample

$\bar{x}$ = the average of the values of x within a sample

$\bar{y}$ = the average of the values of y within a sample





Step 3.4: Mối quan hệ giữa các features

(Multi)collinearity

- **Mức độ 2 hay nhiều features quan hệ tuyến tính với nhau**
- **Thông thường hệ số tương quan > 0.7 hoặc < -0.7 biểu thị rằng 2 hay nhiều features có hiện tượng (đa) cộng tuyến với nhau**

